RACE
Regular Analysis Through Continuous Exercise

Periodic Table : RACE-01

Niels Bohr's classification; Some commonly used terms

1. Which of the following set of atomic numbers represent representative element

(A) 5, 13, 30, 53 (B) 11, 33, 58, 84 (C) 5, 17, 21, 31 (D) 9, 31, 53, 83

CPT218

2. Which of the following pairs shows diagonal relationship ?

(A) Li and Mg (B) Na and K (C) Zn and Cd (D) Li and Be

CPT094

3. Which element are not called transition element ?

(A) Fe, Co, Ni (B) W, Cu, Au (C) Zn, Hg, Cd (D) Sc, V, Mn

CPT096

Long Form of Periodic Table

4. 4th shell of atom contains 4s, 4p, 4d and 4f subshell, which subshells are not filled in any 4th period element?

(A) 4p, 4d, 4f (B) 4s, 4p (C) 4p, 4f (D) 4d, 4f

CPT219

Electronic Configuration of Elements and the periodic table, Electronic Configuration in Period; Groupwise Electronic Configuration, Identification of group, block and period of an element

Bankers Academy JEE

5. In modern periodic table, the element with atomic number Z = 118 will be :

(A) Uuo; Ununoctium; alkaline earth metal (B) Uno; Unniloctium; transition metal

(C) Uno; Unniloctium; alkali metal (D) Uuo; Ununoctium; noble gas

CPT220

Definition of Effective Nuclear Charge (Zeff); Shielding Effect; Penetration Power

6. In Sodium atom the screening is due to:-

(A) $3s^2, 3p^6$ (B) $2s^1$ (C) $1s^2, 2s^2, 2p^6$ (D) $1s^2, 2s^2$

CPT099

7. Screening effect is not observed in:-

(A) $He^{\oplus}$ (B) $Li^{2\oplus}$ (C) H (D) All of these

CPT101

Calculation of Shielding Constant using Slater's Rule

8. Z_{eff} for 4s electron in $_{30}Zn$ is :

(A) 1.65 (B) 4 (C) 12.85 (D) 4.35

CPT105

Ionic Radius; Factors Affecting Ionic Radius Ionic Radius Order for Isoelectronic and Isoprotonic Species

9. In the isoelectronic species, the ionic radii (Å) of N^{3-}, Mg^{2+} and Al^{3+} are respectively given by : -

(A) 1.36, 1.40, 1.71 (B) 1.36, 1.71, 1.40 (C) 1.71, 1.40, 1.36 (D) 1.71, 1.36, 1.40

CPT221

For More Join: @IITJEE_Advanced

Atomic Radius ; Trends for d- and f- Block Elements

10. Which of the following pair has approximately the same atomic radii ?

(A) Zr and Hf (B) Al and Ga (C) (a) and (b) both (D) Al and Mg

CPT109

11. Which of the following orders of atomic / Ionic radius is correct ?

(A) B < Al ≈ Ga (B) Sc > Zn > Cu (C) C < O < N (D) $Al^{3\oplus} < Al^{2\oplus} < Al^{\oplus}$

CPT222

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	A	C	D	D	C	D	D	C	C
Question	11									
Answer	ABD									

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Periodic Table : RACE-02

Diagonal Trends in Ionic Radius

1. Which of the following sequence is correct for decreasing order of ionic radius ?

(A) $Se^{2\ominus}$, $I^{\ominus}$, $Br^{\ominus}$, $O^{2\ominus}$, $F^{\ominus}$ (B) $I^{\ominus}$, $Se^{2\ominus}$, $O^{2\ominus}$, $Br^{\ominus}$, $F^{\ominus}$

(C) $Se^{2\ominus}$, $I^{\ominus}$, $Br^{\ominus}$, $F^{\ominus}$, $O^{2\ominus}$ (D) $I^{\ominus}$, $Se^{2\ominus}$, $Br^{\ominus}$, $O^{2\ominus}$, $F^{\ominus}$

CPT115

Definition of Ionization Energy; Successive Ionization Energy

2. Successive ionisation energies of an element 'X' are given below (in kcal/mol)

IP_1	IP_2	IP_3	IP_4
65	95	556	595

Electronic configuration of the element 'X' is:-

(A) $1s^2, 2s^2\,2p^6, 3s^2\,3p^2$ (B) $1s^2, 2s^1$

(C) $1s^2, 2s^2\,2p^2$ (D) $1s^2, 2s^2\,2p^6, 3s^2$

CPT223

3. The first five ionization energies of an element are 9.1, 16.2, 24.5, 35 and 205.7 eV respectively. Then number of valence electron in the atom is

(A) 2 (B) 3 (C) 4 (D) 5

Pankers Academy JEE

CPT132

Factors Affecting Ionization Energy

4. Correct order of first ionisation energy is :-

(a) Li < B < Be < C (b) O < N < F (c) Be < N < Ne

(A) a, b (B) b, c (C) a, c (D) a, b, c

CPT118

Periodicity of Ionization Energy in s and p Block Elements

5. The decreasing order of the ionization potential of the following elements is :

(A) Ne > Cl > P > S > Al > Mg (B) Ne > Cl > P > S > Mg > Al

(C) Ne > Cl > S > P > Mg > Al (D) Ne > Cl > S > P > Al > Mg

CPT120

6. Decreasing ionization potential for K, Ca & Ba is

(A) Ba > K > Ca (B) Ca > Ba > K

(C) K > Ba > Ca (D) K > Ca > Ba

CPT126

Comparison of IE_3 and IE_4

7. The third ionisation energy is maximum for :

(A) Nitrogen (B) Phosphorous (C) Aluminium (D) Boron

CPT130

For More Join: @IITJEE_Advanced

Ionisation Energy Order for Isoelectronic and Isoprotonic Species

8. Select the incorrect order of I.E.:-

(A) $Cl^{\ominus} > Cl > Cl^{\oplus}$ (B) $Cl^{\oplus} > Cl > Cl^{\ominus}$

(C) $Cl > Cl^{\oplus} > Cl^{\ominus}$ (D) $Cl^{\ominus} > Cl^{\oplus} > Cl$

CPT121

Applications of Ionization Energy; Electropositive Character; Metallic and Non Metallic Nature; Reducing Power and reactivity of metals

9. In which of the following set of elements 1st element is more metallic than second.

(A) Ba, Ca (B) Sb, Sn (C) Ge, S (D) Na, F

CPT141

10. In the periodic table, the metallic character of element : -

(A) decreases from left to right across a period and on descending a group

(B) decreases from left to right across a period and increases on descending a group

(C) increases from left to right across a period and on descending a group

(D) increases from left to right across a period and decreases on descending a group

CPT224

Periodicity of Electron Affinity in s and p block elements

11. The electron affinity values for the halogens show the following trend:

(A) Cl > F > Br > I (B) F < Cl < Br < I (C) F > Cl > Br > I (D) F > Br > Cl > I

CPT225

12. The order of electron gain enthalpy (magnitude) of O, S and Se is :

(A) O > S > Se (B) S > Se > O (C) Se > S > O (D) S > O > Se

CPT226

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	D	C	D	B	B	A	ACD	ACD	B
Question	11	12								
Answer	A	B								

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Periodic Table : RACE-03

2nd Electron Affinity

1. Process in which maximum energy is released :-

(A) $O \rightarrow O^{2\ominus}$ (B) $Mg^{\oplus} \rightarrow Mg^{2\oplus}$ (C) $Cl \rightarrow Cl^{\ominus}$ (D) $F \rightarrow F^{\ominus}$

CPT134

2. The formation of the oxide ion O^{2-} (S) requires first an exothermic and then an endothermic step as shown below :

$$O(g) + e^{\ominus} \rightarrow O^{\ominus}(g) ; \quad \Delta H^{\circ} = -142 \text{ kJ mol}^{-1}$$

$$O^{\ominus}(g) + e^{\ominus} \rightarrow O^{2\ominus}(g) ; \quad \Delta H^{\circ} = +744 \text{ kJ mol}^{-1}$$

This is because:

(A) Oxygen is more electronegative

(B) Oxygen has high electron affinity

(C) $O^{\ominus}$ ion has comparatively larger size than oxygen atom

(D) $O^{\ominus}$ ion will tend to resist the addition of electron

CPT227

Factors Affecting Electronegativity; Periodicity of Electronegativity in s-p- block elements

3. The electronegativities of the following elements: H, O, F, S and Cl increase in the order :-

(A) H < O < F < S < Cl (B) Cl < H < O < F < S

(C) H < S < O < Cl < F (D) H < S < Cl < O < F

CPT146

Different Scales to Measure Electronegativity

4. Which of the following relation is correct ?

(A) 2 IP – EA – EN = 0 (B) 2 IP – EN + EA = 0

(C) 2 EN – IP – EA = 0 (D) EN – IP – EA = 0

CPT150

Applications of Electronegativity; Polarity of Bond; Calculation of Percentage Ionic Character, Effect of EN on bond length

5. X-X bond length is 1.00Å and C-C bond length is 1.54Å. If electronegativities of X and C are 3.0 and 2.0 respectively then C-X bond length is likely to be ?

(A) 1.11 (B) 2.1Å (C) 1.18Å (D) 2.4Å

CPT111

6. Among the following least and most polar bonds are respectively:-

(a) C – I (b) N – O (c) C – F (d) P – F

(A) d and c (B) a and d (C) b and d (D) b and c

CPT147

Applications of Electronegativity; Acidic Strength of Oxides and Hydroxides

7. The most basic among these hydroxides is :

(A) $Be(OH)_2$ (B) $Mg(OH)_2$ (C) $Ca(OH)_2$ (D) $Ba(OH)_2$

CPT167

For More Join: @IITJEE_Advanced

8. The increasing order of acidic nature of Li_2O, BeO, B_2O_3

(A) $Li_2O > BeO < B_2O_3$ (B) $Li_2O < BeO < B_2O_3$

(C) $Li_2O < BeO > B_2O_3$ (D) $Li_2O > BeO > B_2O_3$

CPT170

Types of Oxides

9. Which one of the following oxides is neutral ?

(A) CO (B) SnO_2 (C) ZnO (D) SiO_2

CPT168

Acidic Strength of Hydra Acids

10. Which is incorrect order ?

(A) $H_2O < H_2S < H_2Se < H_2Te$ (acidic nature) (B) $H_2O < H_2S < H_2Se < H_2Te$ (basic nature)

(C) HF < HCl < HBr < HI (acidic nature) (D) HF > HCl > HBr > HI (basic nature)

CPT169

11. Select correct order for acidic strength of hydra acid

(A) $HF > H_2O$ (B) HF > HBr (C) $H_2O > NH_3$ (D) $HF > NH_3$

CPT172

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	D	D	C	C	B	D	B	A	B
Question	11									
Answer	ACD									

For More Join: @IITJEE_Advanced